

DEPI Graduation Project

Project Planning & Management

System Under Test: Swag Labs (SauceDemo)

<https://www.saucedemo.com>

Date: April 23, 2026

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1.1 Project Proposal

Overview

This project aims to deliver a comprehensive quality assurance evaluation of the Swag Labs web application (<https://www.saucedemo.com>), an e-commerce demo platform provided by Sauce Labs. The evaluation covers manual functional testing, REST API testing (where applicable), and Selenium-based test automation using Java, TestNG, and the Page Object Model (POM) design pattern.

Objectives

- Identify functional defects across the core user journeys: Authentication, Product Browsing, Cart Management, and Checkout.
- Validate non-functional attributes including UI consistency, performance under normal load, and security basics.
- Produce structured, traceable test documentation compliant with the DEPI graduation project guidelines.
- Deliver an automation suite (Selenium + TestNG + Java) covering at least 70% of identified functional test cases.
- Document all discovered bugs with severity/priority classification and reproducible steps.

Scope

In Scope:

- Login / Logout (all provided user types)
- Product inventory listing, sorting, and filtering
- Product detail page
- Shopping cart (add, remove, persist)
- Checkout flow (Step 1: Info → Step 2: Overview → Order Confirmation)
- Navigation menu (reset app state, about page redirect)
- UI/UX validation

Out of Scope:

- Back-end / database direct access (no source code access)
- Payment gateway integration (none exists in demo)
- Mobile native application testing

1.2 Project Plan – Timeline & Milestones

Phase	Deliverable	Graduate Deadline	Student Deadline	Status
Phase 1 – Planning	Project Proposal, Plan, Risk Assessment	27-Jan-2026	20-Feb-2026	Completed
Phase 2 – Literature	Feedback, Grading Criteria	01-Mar-2026	20-Apr-2026	Completed
Phase 3 – Requirements	User Stories, FR/NFR, Use Cases	01-Mar-2026	20-Apr-2026	Completed
Phase 4 – Analysis	ERD, DFD, Architecture, Wireframes	03-Apr-2026	01-May-2026	Completed
Phase 5 – Implementation	Source Code, Automation Suite, README	17-May-2026	10-Jul-2026	In Progress
Phase 6 – Final Delivery	Test Report, Bug Report, Presentation	24-May-2026	17-Jul-2026	Pending

1.3 Task Assignment & Roles

Role	Responsibilities
QA Lead / Project Owner	Test strategy, planning, deliverable oversight, final report
Manual Tester	Exploratory testing, test case authoring, bug reporting
Automation Engineer	Selenium POM framework setup, TestNG suite execution, CI integration
Business Analyst	Requirements gathering, use-case modeling, stakeholder analysis
Technical Writer	User manual, API documentation, presentation deck

1.4 Risk Assessment & Mitigation Plan

Risk	Likelihood	Impact	Mitigation
SauceDemo site unavailability	Low	High	Cache test data; document steps offline; use screenshots.
Scope creep in test coverage	Medium	Medium	Strictly follow approved test plan; log deferred items as out-of-scope.
Automation flakiness (dynamic content)	Medium	High	Implement explicit waits; retry logic in TestNG; stable selectors via POM.
Insufficient time for full coverage	Medium	Medium	Prioritize smoke + regression test cases; manually test lower-priority cases.
Credential/user behavior changes	Low	Medium	Document all known user types upfront; add assertions for error message validation.
Toolchain compatibility issues	Low	Low	Pin dependency versions in Maven pom.xml; use WebDriverManager.

1.5 Key Performance Indicators (KPIs)

KPI	Target	Measurement Method
Test Case Coverage	≥ 90% of identified functional requirements	TC Executed / TC Planned × 100
Defect Detection Rate	≥ 80% detected before final delivery	Bugs Found / Total Bugs × 100
Automation Coverage	≥ 70% of regression test cases automated	Automated TC / Total Regression TC
Test Execution Pass Rate	≥ 85% on final regression run	Pass / Total Executed × 100
Defect Resolution Rate	100% Critical/High bugs documented	Resolved + Documented / Total Reported
Documentation Completeness	All required DEPI sections submitted	Checklist-based review